Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning and AI Engineer with **3 years** of industry experience building **production GenAI and data systems for an investment firm** (**HDFC Asset Management**): **RAG applications and tool-calling agents** used by Compliance, Client Service and Investment teams, and a **Morningstar** fund-data platform. Designed a **scoring and ranking algorithm** returning the evidence behind each result in an **AstraZeneca** dissertation for an **MSc in Data Science at the University of Bristol** (awaiting result); builds with **AI-assisted coding** (**Claude Code**). Seeking the **Machine Learning and AI Engineer** role at **Reflexivity**, building AI systems that turn financial data into investor insight.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production GenAI & Agentic Systems

- Designed and built **production GenAI** applications end to end (**ReactJS**, **Python FastAPI** microservices, **LlamaIndex**, **Azure OpenAI GPT-3.5 & Llama-3**) for Compliance, Client Service and Investment, initially as the first engineer on the team, with a separate **pgvector** database per department.
- Built a **WhatsApp** fund-comparison bot (via the **Netcore** API) used daily by the Sales team (roughly **50 to 100 users**, **400 to 1,000 requests** in working hours): **Azure OpenAI tool calling** picks the funds and comparison type and calls **Python functions** over SQL data, returning fund indicators and a chart PDF. Parameter extraction broke first; fixed with example inputs in the **JSON tool definitions**.
- Took applications from prototype to real users, iterating on what broke; guardrailed behaviour through **system prompts**, versioned every **prompt change in Git** and had **Compliance** review each release; allocated models by cost, **self-hosting Llama-3 on a GPU EC2 instance** for the largest team.

Investment Data & Signal Pipelines

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies, fuzzy-matches them to the companies analysts track (dropping the rest) and sends a daily summary of previous and changed ratings, saving **60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when an agency began sending images.
- Analyzed customer data with **Python Pandas** and **Market Basket Analysis (Apriori Algorithm)** to surface buying patterns for the marketing team; built **Power BI** dashboards for the investment team's fortnightly credit report analysis, moving analysts from Excel and improving efficiency by **10 hours a week**.
- Worked directly with **Investment, Compliance and Client Service** stakeholders to turn requirements into working tools; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

MLOps, Deployment & Reliability

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS (EC2 including a GPU instance, S3, ECR and Lambda**), packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Ran both **real-time and batch workloads**: user-facing services on FastAPI microservices, and a daily rating-email pipeline run on **AWS SageMaker** and scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **scoring and ranking algorithm** that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages; the **data preprocessing pipeline** that builds 14 clean, joined tables from 17 raw inputs with integrity checks; and the **RAG-based AI assistant**. Returns a ranked shortlist with the evidence behind each pick and a confidence tier, refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM with the model selectable per request). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; rate limits shaped the design (12 calls instead of 108, results streamed). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with LSTM/attention decoders on three synthetic datasets across five seeds, evaluated with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolutional Neural Network (CNN)** model.)

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework for AI-assisted coding) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite